CONNECTED HOME ENERGY FORUM · BERLIN, 26 OCTOBER 2026

Electrification without interoperability

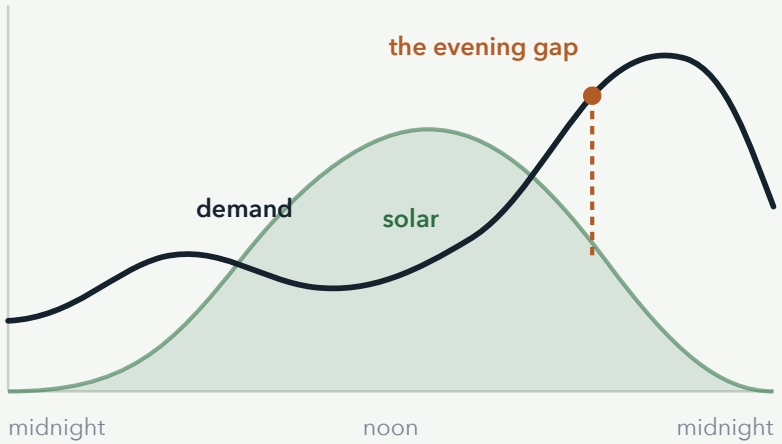
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Electrification creates two problems. Today we are solving one.

BALANCE THE GRID

The entire grid, hour by hour

Supply must match demand, always.

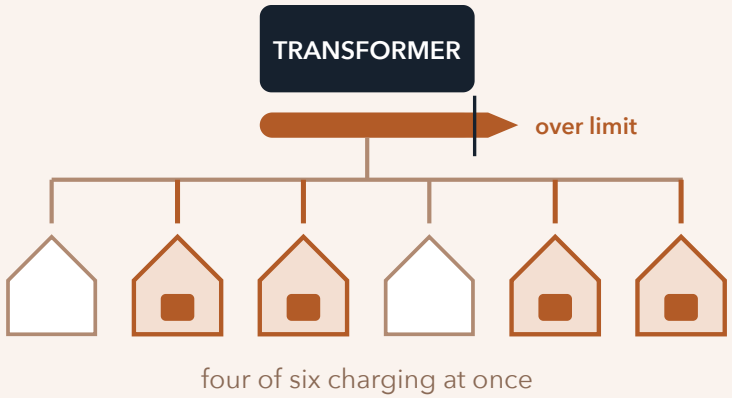


An economics problem. Move energy in time.

PROTECT THE TRANSFORMER

One block, right now

The transformers and wires have a fixed capacity.



four of six charging at once

A physics problem. Cap power right now.

Balancing the grid is an **economics** problem. Protecting the transformer is a **physics** problem. **Both land on the customer's bill.** Peak we cannot shift is generation and transmission somebody has to build; a transformer we cannot protect is a street somebody has to dig up.

One signal cannot do both jobs

	A DYNAMIC PRICE	A PER-CUSTOMER POWER LIMIT
What it is	What energy costs right now	How much power this connection may draw right now
Unit	The entire grid	One service connection
What it solves	Balancing the grid	Protecting the transformer
What it does	Moves energy across time	Moves nothing. Caps power, immediately
Without it	Nobody has a reason to shift	Nothing stops everyone shifting at once

A price is an incentive shared by everyone on the same asset. It cannot stop five houses on one transformer responding to the same number at the same moment. **A price is an incentive over a group. A limit is an obligation on one connection.**

WHAT THE DISTRIBUTION NETWORK IS ALREADY DOING

We are already experiencing the transformer problem

Voltage rises on a sunny afternoon, and inverters and chargers trip off to protect themselves. The customer sees a fault, not a grid condition.

Connections are refused, queued or derated for heat pumps and chargers, because nobody can promise what a house will draw at the worst moment.

Capacity is managed by curtailment and by waiting, because there is no signal that tells a connection how much it may take right now.

> Every one of these is about protecting the transformer. **No signal in use today carries an answer to them.**

The good news: three standards already carry a price

	CARRIES A PRICE TODAY
OpenADR 3	Utility to the customer's own equipment, over the Internet
Matter	The Commodity Price cluster, inside the home
EEBUS	Time of Use Tariff: "the absolute price SHALL be included", currency per Wh

Tomorrow you will hear from Matter and from OpenADR on exactly this. **This half of the problem has an answer.**

NOT A PROPOSAL. A RUNNING SERVICE.

A free, public price server, no account required

About 1,700 programs. Real hourly prices for two large utilities across 105 distribution circuits and substations, plus prices computed from published rate schedules.

Hourly marginal carbon, for 11 grid regions, on the same server, in the same shape. A greenhouse-gas number is price-shaped: a device optimizes against it exactly as it does a price.

No authentication, no contract, no enrollment. Open-source home automation receives it and orchestrates the house against it.

> `price.grid-coordination.energy/openadr3/3.1.0/programs?limit=5`
> Try it in a browser during the break.

Three standards already define a connection limit

	A CONNECTION-LEVEL IMPORT/EXPORT LIMIT
OpenADR 3	Yes, as event payloads
EEBUS	**Power Envelope**, defined at "the building's grid connection point", import and export, time-varying, with a declared fallback
Matter	`PowerThresholdStruct` is defined in the Application Cluster specification
S2 / EN 50491-12-2	No site-level construct, and the design FAQ rules out faking one by aggregating devices
HCA Energy Management 1.0	No power quantity of any kind occurs in the specification

The design work is done. The requirement is missing.

This room specified the object years ago. EEBUS defines a Power Envelope at the building's grid connection point, import and export, time-varying, with a declared fallback. That is the right object, defined in the right place, and it predates most of this conversation.

Nothing obliges a device to receive one. Not a standard, not a programme, not a procurement.

What does get fielded caps the enrolled devices rather than the connection. A limit that binds only the appliances in a scheme leaves the transformer exposed to everything else in the house.

> **The gap is not invention. It is obligation.** The objects exist in several of the standards represented in this room. Nothing requires a device to receive one, and nothing certifies that it does.

WHAT HAPPENS WHEN THE INSTRUMENT LEAVES THE INTERFACE OUT OF SCOPE

92.57 per cent

The **EU Code of Conduct on interoperability of energy smart appliances** declares the communication layer out of scope.

The Commission's own JRC evaluation counted **376 registered compliant products**.

HOW THE CAPABILITY IS IMPLEMENTED	SHARE
Manufacturer cloud digital twin	**92.57%**
Built into the appliance	6.90%
Dongle	0.53%

Published by the regulator that wrote the instrument. **Leave the interface out of scope and the market fills it with a cloud you do not control and cannot certify.** The structural answer is a standard that is local by construction, which is what makes tomorrow's Matter session worth your attention.

A CURRENT EVSE CERTIFICATION PROGRAM, SCORED AGAINST SIX TESTS

It certifies dispatch, and neither signal

TEST	VERDICT	IN THE RELEASED SPECIFICATION
1. An objective, not operation	**Not met**	A Charging Schedule Function with default and **dispatch** schedules
2. Open all the way to the appliance	**Met**	Mapped onto an open protocol that terminates on the charger
3. A server the customer chooses	**Not met**	"server address" and "endpoint": zero occurrences
4. Direct from the utility	**Not met**	Aggregator mediated throughout
5. A limit that holds locally	**Not met**	"capacity limit", "import limit", "export limit": zero occurrences
6. The customer who cannot respond	**Met**	A local override, which expires every time the vehicle is unplugged

Yes, and also capacity

Everything you will hear tomorrow is right, and none of it is enough on its own. Matter makes the home coherent and does it locally: one commissioning experience across vendors, a price cluster already in the specification, and no cloud in the middle. OpenADR carries the grid's need to the customer's own equipment. Open source removes the cost of implementing both.

The price half is converging. Four of the standards in this room carry a price today, and one of them is serving real prices to anybody who asks, right now, with no account.

The capacity half has objects and no obligations. It is defined in several specifications, required by none, certified by none, and absent from the marks.

> **This is not a gap in anyone's design. It is a gap in what we ask for.**

ONE ASK

When you write a programme, a tariff or a procurement, ask for both

Ask for a price the device can read, from a server the customer can point somewhere else.

Ask for a connection limit the device must respect, that holds when the network is down.

Then certify that it does. A mark that certifies dispatch and neither signal tells a buyer nothing about whether the grid can be protected.

THE ASK

Nobody in this room has to invent anything to do this. The objects already exist in the standards you already build on. **Put them in the next requirement you write, and the market will deliver them inside two years.**

FOR YOUR TABLES

Four questions I would like your answers to

What is the biggest blocker: the objects, the obligation, or the business model?

Who should own the connection limit? The DNO publishes it, the customer lives under it, the device obeys it, and today no one carries it between them.

What evidence would unlock investment? Name the specific proof your organisation would need.

If a limit arrived at the house tomorrow, what in your product would act on it?

> I am here for both days. **Come and tell me I am wrong.**

EVERYTHING HERE IS PUBLIC AND FREE

Where to find all of this

The price server	`price.grid-coordination.energy/openadr3/3.1.0/programs?limit=5`
User guide, tutorials, client libraries	`github.com/grid-coordination`
The argument, in full	`grid-coordination.energy`
The six tests	`grid-coordination.energy/presentations/`

No account, no contract, no enrollment on any of it.